# Dataset Collection Guide

Public data, your own field data, labelling, and leak-free splits

## 1. Public datasets to download

Check each licence and note the source URL in `dataset_log.csv`. Cite each in the paper.

| Dataset | What it gives us | Use as |
|---|---|---|
| PlantVillage (tomato subset) | Many labelled tomato leaf images over about 10 classes (bacterial spot, early blight, late blight, leaf mold, septoria leaf spot, spider mites, target spot, yellow leaf curl virus, mosaic virus, healthy). Lab-style backgrounds. | Training + in-domain test |
| PlantDoc | Real-world (field/web) plant disease images, includes tomato classes. Names differ slightly from PlantVillage, so build a class-name mapping table. | Cross-domain test |
| Other tomato field sets (search Kaggle / Mendeley Data / IEEE DataPort) | Extra field images. Verify licence, size, and label quality before using. | Optional extra training / test |
| Open Images / COCO subsets | 'Plant', 'houseplant', 'tree' boxes for the detector; also people/background images as hard negatives. | Detector training |

**TIP:** Do not buy datasets. Free ones are enough. If a dataset has no clear licence, do not use it in the paper.

## 2. Collecting your own tomato images (field or potted plants)

Even 300-500 good images per class from your own camera make the paper far more convincing, because they test the real hardware under real conditions.

### Before you go

[ ] Permission from the farm owner (ask politely; explain it is for a college project)

[ ] Phone fully charged; ESP32-CAM bench kit with power bank; the capture script tested at home

[ ] A notebook or phone notes app for plant IDs; plant tags (paper tags, tape, marker)

[ ] If possible, an agriculture officer / KVK expert / experienced farmer who can confirm which plants are diseased

### Plant ID system (essential for leak-free splits)

Tag every plant with a unique ID such as **T-001, T-002, ...**. Take all photos of one plant under that ID. Record whether the plant is healthy or which disease it has, confirmed by the expert.

### Photo protocol per plant (15-30 photos)

- Distances: 15 cm, 30 cm, 50 cm. Angles: front, left, right, top, and leaf underside when relevant.
- Different parts: young leaves, old leaves, stem, fruit if visible.
- Lighting: take some in direct sun, some in shade; do not skip hard light, it is the real condition.
- Include natural backgrounds (soil, other leaves, mud). Do not hold a white sheet behind leaves.
- Use the **ESP32-CAM** for at least half the photos and your phone for the rest; record the source in metadata.
- Take a few photos of **non-plant scenes** (soil, pots, people's legs, fence, tools) for the detector's hard negatives.
- Avoid blurry photos; delete obvious failures on the spot.

### Run the capture script

```
python "Starter Code/capture_dataset.py" http://<camera-ip>
```

It asks for plant ID, label, weather and distance, then saves each photo and one metadata row. For phone photos, fill the same columns by hand in `Templates/dataset_log.csv`.

## 3. Labelling rules

| Label | Use when | Who confirms |
|---|---|---|
| healthy | No visible lesions, spots, curling, discolouration | Expert + you |
| early_blight / late_blight / ... (disease name) | Expert identifies the disease | Expert |
| uncertain (do NOT train on it) | Not sure; could be nutrient stress, pest or damage | Move to a separate folder |

The word 'unhealthy' hides many causes (nutrient deficiency, pests, sunburn, disease). For the paper we use **specific disease names** and derive the binary healthy/diseased label from them.

## 4. Splitting: by plant, never by image

If 20 photos of one leaf end up half in training and half in testing, the model recognises the leaf, not the disease, and accuracy looks falsely perfect. `make_splits.py` assigns whole plants to train, val or test (about 70/15/15). For the public datasets, which lack plant IDs, hash near-duplicates or at least split by source file group; state this limitation in the paper.

```
python "Starter Code/make_splits.py"
```

Afterwards verify: no plant_id appears in two splits (the script prints counts; also check by hand). Keep the final test set locked: look at it only for the final evaluation.

## 5. Target numbers (guidance, not requirements)

| Item | Minimum to report | Good |
|---|---|---|
| Own field images per class (healthy + 2-3 diseases) | 100 | 300-500 |
| Distinct own plants per class | 5 | 10-20 |
| Detector training images (plants + negatives) | 1,000 | 3,000+ |
| Public tomato images for classifier training | all available | all available |

If you cannot reach the minimum for your own set, say so plainly in the results and treat that test as a pilot.

## 6. Data hygiene and ethics

- Do not photograph people's faces; delete such images. Do not include farm names or locations in anything that goes into the paper.
- Back up the raw data to a Drive folder weekly. Never edit raw files; process copies.
- Keep a file `DATA_SOURCES.txt` listing every dataset, URL, licence and download date.